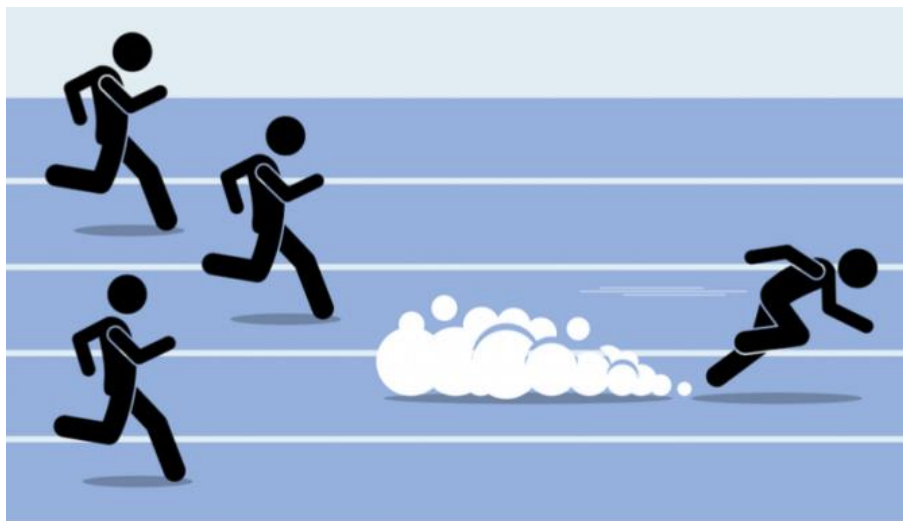


Name: _____ Period: _____

AP Chem Unit 3: Kinetics



Textbook Reference: Zumdahl Chemistry: A Molecular Approach Chapter 12

Date	Day	Learning Targets	Activities	Work on this!
9/24-25	U3D1	What is spectroscopy?	- Review Exam 2 - Overview: Spectroscopy	
9/26-29	U3D2	What is a reaction rate?	- Lab Simulation: Beer's Law - Notes: Rate Expressions	Finish Beer's Law Lab Start Textbook Probs?
9/30-10/1	U3D3	What is a differential rate law?	- Differential Rate Law	Work through pg. 7 of your packet
10/2-3	U3D4	What is an integrated rate law?	- Quiz? - Practice FRQ - Notes and practice: Integrated Rate Laws	
10/6-7	U3D5	What is a reaction mechanism?	- The Great Parent Note Forgery Caper - Reaction Mechanisms	
10/8-9	U3D6	How are mechanisms and rate laws related?	QUIZ	Textbook set #1 due
10/10-14	U3D7	How are rate laws measured in the lab?	Lab	
10/15-16	U4D1	How are rate laws and equilibrium related?	- Review - Penny Ante Equilibrium	
10/17-21	U3D8	What do you know about kinetics?	Unit 3 Exam	All Textbook Problems due

Unit 3: Chemical Kinetics	
3.11	Spectroscopy and the Electromagnetic Spectrum
3.11A	Explain relationship between a region of the electromagnetic spectrum and types of molecular or electronic transitions associated with that region.
	3.11.A.1: Differences in absorption or emission of photons in different spectral regions are related to the different types of molecular motion or electronic transition: Microwave radiation is associated with transitions in molecular rotational levels. Infrared radiation is associated with transitions in molecular vibrational levels. Ultraviolet / visible radiation is associated with transitions in electronic energy levels.
3.13	Beer-Lambert Law

3.13A	Explain the amount of light absorbed by a solution of molecules or ions in relationship to the concentration, path length, and molar absorptivity.
	3.13.A.1: The Beer-Lambert law relates the absorption of light by a solution to three variables according to the equation: EQN: $A = \epsilon bc$.
	The molar absorptivity, ϵ , describes how intensely a chemical species absorbs light of a specific wavelength. The path length, b , and concentration, c , are proportional to the number of light-absorbing particles in the light path.
	3.13.A.2: In most experiments the path length and wavelength of light are held constant. In such cases, the absorbance is proportional only to the concentration of absorbing molecules or ions. The spectrophotometer is typically set to the wavelength of maximum absorbance (optimum wavelength) for the species being analyzed to ensure the maximum sensitivity of measurement.
5.1	Reaction Rates
5.1A	Explain the relationship between the rate of a chemical reaction and experimental parameters.
	5.1.A.1: The kinetics of a chemical reaction is defined as the rate at which an amount of reactants is converted to products per unit of time.
	5.1.A.2: The rates of change of reactant and product concentrations are determined by the stoichiometry in the balanced chemical equation.
	5.1.A.3: The rate of a reaction is influenced by reactant concentrations, temperature, surface area, catalysts, and other environmental factors.
5.2	Introduction to Rate Law
5.2A	Represent experimental data with a consistent rate law expression.
	5.2.A.1: Experimental methods can be used to monitor the amounts of reactants and / or products of a reaction and to determine the rate of the reaction.
	5.2.A.2: The rate law expresses the rate of a reaction as proportional to the concentration of each reactant raised to a power.
	5.2.A.3: The power of each reactant in the rate law is the order of the reaction with respect to that reactant. The sum of the powers of the reactant concentrations in the rate law is the overall order of the reaction.
	5.2.A.4: The proportionality constant in the rate law is called the rate constant. The value of this constant is temperature dependent and the units reflect the overall reaction order.
	5.2.A.5: Comparing the initial rates of a reaction is a method to determine the order with respect to each reactant.
5.3	Concentration Change vs Time
5.3A	Identify the rate law expression of a chemical reaction using data that show how the concentrations of reaction species change over time.
	5.3.A.1: The order of a reaction can be inferred from a graph of concentration of reactant versus time.
	5.3.A.2: If a reaction is first order with respect to a reactant being monitored, a plot of the natural log (ln) of the reactant concentration as a function of time will be linear.
	5.3.A.3: If a reaction is second order with respect to a reactant being monitored, a plot of the reciprocal of the concentration of that reactant versus time will be linear.
	5.3.A.4: The slopes of the concentration versus time data for zeroth, first, and second order reactions can be used to determine the rate constant for the reaction. Zeroth Order EQN: $[A]_t - [A]_0 = -kt$ First Order EQN: $\ln[A]_t - \ln[A]_0 = -kt$ Second Order EQN: $1/[A]_t - 1/[A]_0 = kt$
	5.3.A.5: Half-life is a critical parameter for first order reactions because the half-life is constant and related to the rate constant for the reaction by the equation: EQN: $t_{1/2} = 0.693/k$.
	5.3.A.6: Radioactive decay processes provide an important illustration of first order kinetics.
5.4	Elementary Reactions
5.4A	Represent an elementary reaction as a rate law expression using stoichiometry.
	5.4.A.1: The rate law of an elementary reaction can be inferred from the stoichiometry of the particles participating in a collision.
	5.4.A.2: Elementary reactions involving the simultaneous collisions of three or more particles are rare.
5.5	Collision Model
5.5A	Explain the relationship between the rate of an elementary reaction and the frequency, energy, and orientation of particle collisions.
	5.5.A.1: For an elementary reaction to successfully produce products, reactants must successfully collide to initiate bond-breaking and bond-making events.
	5.5.A.2: In most reactions, only a small fraction of the collisions lead to a reaction. Successful collisions have both sufficient energy to overcome the activation energy requirements and orientations that allow the bonds to rearrange in the required manner.

	5.5.A.3: The Maxwell-Boltzmann distribution curve describes the distribution of particle energies; this distribution can be used to gain a qualitative estimate of the fraction of collisions with sufficient energy to lead to a reaction, and also how that fraction depends on temperature.
5.6	Reaction Energy Profile
5.6A	Represent the activation energy and overall energy change in an elementary reaction using a reaction energy profile.
	5.6.A.1: Elementary reactions typically involve the breaking of some bonds and the forming of new ones.
	5.6.A.2: The reaction coordinate is the axis along which the complex set of motions involved in rearranging reactants to form products can be plotted.
	5.6.A.3: The energy profile gives the energy along the reaction coordinate, which typically proceeds from reactants, through a transition state, to products. The energy difference between the reactants and the transition state is the activation energy for the forward reaction.
	5.6.A.4: The rate of an elementary reaction is temperature dependent because the proportion of particle collisions that are energetic enough to reach the transition state varies with temperature. The Arrhenius equation relates the temperature dependence of the rate of an elementary reaction to the activation energy needed by molecular collisions to reach the transition state.
5.7	Introduction to Reaction Mechanisms
5.7A	Identify the components of a reaction mechanism.
	5.7.A.1: A reaction mechanism consists of a series of elementary reactions, or steps, that occur in sequence. The components may include reactants, intermediates, products, and catalysts.
	5.7.A.2: The elementary steps when combined should align with the overall balanced equation of a chemical reaction.
	5.7.A.3: A reaction intermediate is produced by some elementary steps and consumed by others, such that it is present only while a reaction is occurring.
	5.7.A.4: Experimental detection of a reaction intermediate is a common way to build evidence in support of one reaction mechanism over an alternative mechanism.
5.8	Reaction Mechanism and Rate Law
5.8A	Identify the rate law for a reaction from a mechanism in which the first step is rate limiting.
	5.8.A.1: For reaction mechanisms in which each elementary step is irreversible, or in which the first step is rate limiting, the rate law of the reaction is set by the molecularity of the slowest elementary step (i.e., the rate-limiting step). ★
5.9	Pre-Equilibrium Approximation
5.9A	Identify the rate law for a reaction from a mechanism in which the first step is not rate limiting.
	5.9.A.1: If the first elementary reaction is not rate limiting, approximations (such as pre-equilibrium) must be made to determine a rate law expression.
5.10	Multistep Reaction Energy Profile
5.10A	Represent the activation energy and overall energy change in a multistep reaction with a reaction energy profile.
	5.10.A.1: Knowledge of the energetics of each elementary reaction in a mechanism allows for the construction of an energy profile for a multistep reaction.
5.11	Catalysis
5.11A	Explain the relationship between the effect of a catalyst on a reaction and changes in the reaction mechanism.
	5.11.A.1: In order for a catalyst to increase the rate of a reaction, the addition of the catalyst must increase the number of effective collisions and / or provide a reaction path with a lower activation energy relative to the original reaction coordinate.
	5.11.A.2: In a reaction mechanism containing a catalyst, the net concentration of the catalyst is constant. However, the catalyst will frequently be consumed in the rate-determining step of the reaction, only to be regenerated in a subsequent step in the mechanism.
	5.11.A.3: Some catalysts accelerate a reaction by binding to the reactant(s). The reactants are either oriented more favorably or react with lower activation energy. There is often a new reaction intermediate in which the catalyst is bound to the reactant(s). Many enzymes function in this manner.
	5.11.A.4: Some catalysts involve covalent bonding between the catalyst and the reactant(s). An example is acid-base catalysis, in which a reactant or intermediate either gains or loses a proton. This introduces a new reaction intermediate and new elementary reactions involving that intermediate.
	5.11.A.5: In surface catalysis, a reactant or intermediate binds to, or forms a covalent bond with, the surface. This introduces elementary reactions involving these new bound reaction intermediate(s).

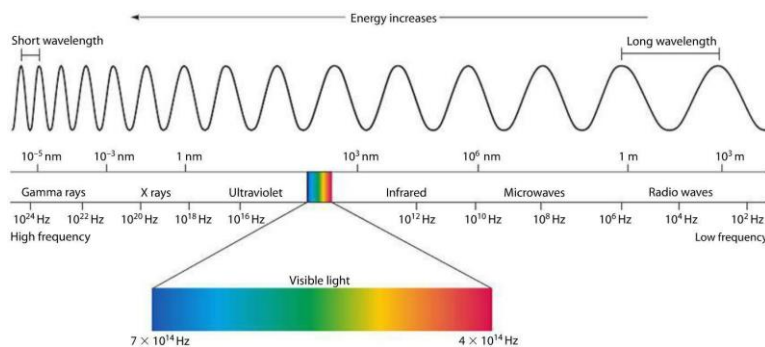
AP Chem Spectroscopy and Beer's Law Notes

3.11 Spectroscopy and the Electromagnetic Spectrum

3.13 Beer-Lambert Law

Spectroscopy

- Spectroscopy** is the study of how matter interacts with electromagnetic radiation. Electromagnetic radiation is classified into different types based on the wavelength (or frequency) of the radiation.



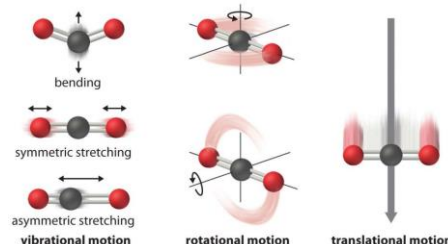
Molecular Movement

Molecules can move in different ways. These can be categorized as:

_____ - movement from one place to another

_____ - spinning in place

_____ - moving in place, the movement of the atoms within the molecule relative to one another, such as bending, stretching and other internal movements within the molecule.



Microwave Rotational Spectroscopy

- Microwave radiation is _____ in energy than visible light. When the microwave radiation strikes the particles it causes them to rotate. This is due to the interaction of the dipole of the molecules interacting with the electromagnetic field of the microwave photons.
- Key Point: Microwave radiation is associated with transitions in molecular _____ levels.**

Infrared Vibrational Spectroscopy

- Infrared spectroscopy is _____ in energy than microwaves, but _____ than visible light. Infrared (IR) spectroscopy measures the vibration of the atoms and allows the determination of the functional groups present in a molecule.
- It analyzes absorption, emission and reflection. Generally, lighter atoms and stronger bonds tend to vibrate at higher frequencies.
- Key Point: Infrared radiation is associated with transitions in molecular _____ levels.**

Ultraviolet-Visible (UV-Vis) Light Spectroscopy

- The interaction between visible and UV wavelengths of electromagnetic spectrum and molecules or atoms can allow the determination of the _____ of the molecules in the solution. The photons of colored light are absorbed by a compound causing electrons to move from their ground state to a higher energy “excited” state. By measuring the amount of light leaving (transmitted from) a sample and comparing it to the amount of light that entered the sample we can find the amount of light that was absorbed by the sample. This is completed using a device called a spectrophotometer. By knowing the amount of light absorbed, one can determine the concentration of a colored substance within a solution.
- Key Point: Ultraviolet/visible radiation is associated with transitions in _____ energy levels. It analyzes absorption, emission and reflection.

PRACTICE

Match the type of electromagnetic radiation with the type of molecular motions or transitions that it causes.

A) Infrared	Rotational Motion
B) Ultraviolet-Visible Light	Electronic Transitions
C) Microwaves	Vibrational Motion

Two samples of copper (II) sulfate were analyzed using visible light spectroscopy. Sample One absorbed twice as much light as Sample Two.

- a) What can you deduce about the concentrations of the two solutions?
- b) What must be true of the solutions in order to get results from this type of spectroscopy?
- c) Describe the type of changes that occur within the molecule during the process of UV-Vis spectroscopy.

Colorimetry

- Colorimetry is the use of the human eye to determine the concentration of _____ species.
- Spectrophotometry is the use of _____ to make the same measurements. It extends the range of possible measurements beyond those that can be determined by the eye alone.
- Visual Observations – Because colorimetry is based on inspection of materials with the human eye, it is necessary to review aspects of visible light.
- Visible light is the narrow range of electromagnetic waves with the wavelength of _____ nm.

- Complementary Colors lie across the diameter on the color wheel and combine to form “white light”, so the color of a compound seen by the eye is the _____ of the color of light absorbed by a colored compound; thus it completes the color.
- Spectrophotometer** - an instrument that measures the _____ of light absorbed, or the intensity of color at a given wavelength.
- The _____ **of color** can be given a numerical value by comparing the amount of light prior to passing it through the sample and after passing through the sample.
- These **quantitative measurements** of light absorbed are the _____ and the _____. In AP Chem, we only will consider _____
- Transmittance- how much light _____ through the sample
- Absorbance- how much light the sample _____; proportional to the _____ of the solution

Beer Lambert Law

Beer- Lambert Law (aka Beer’s Law)- the linear relationship between _____ and _____ of an absorbing species.

$$A = bcd$$

$$\text{OR in AP Chem World..... } A = \epsilon bc$$

A is the absorbance- amount of _____ that was absorbed

“b” is molar absorptivity in $L/[(\text{mole})(\text{cm})]$ - _____ to the analyte

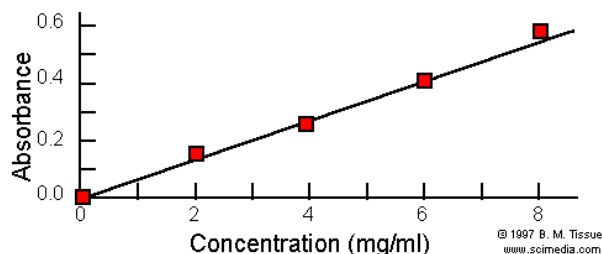
“c” is the _____ of the analyte (sample) in mol/L

“d” is the _____ (distance) in cm that the light traveled through the sample (almost always _____ cm)

There is a _____ proportionality between absorbance and the concentration of a solution.

A blue solution of copper (II) sulfate has its absorbance measured and compared to a plot of standardized solutions.

- What color wavelength is recommended for the spectroscopy?
- What is the molar concentration of a solution with absorbance 0.35?



AP Chem Kinetics Notes

5.1: Reaction Rates

5.2 Introduction to Rate Law

The rate of a reaction can be measured

Reaction Rate- change in _____ of a reactant or product per unit _____.

If $[A] =$

$$\text{Rate} = \frac{\Delta[A]}{\Delta t}$$

The above gives the _____ rate over a period of time

Reaction Rate can be positive or negative

Consider the reaction: $A + 2B \rightarrow C$

- If the rate expression involves a _____, then the concentration is _____:

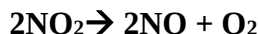
$$\text{Rate} = - \Delta[A] / \Delta t$$

Rate is expressed as _____ because $[A]$ decreases

- If the rate expression involves a _____, then the concentration is _____:

$$\text{Rate} = +\Delta[C] / \Delta t$$

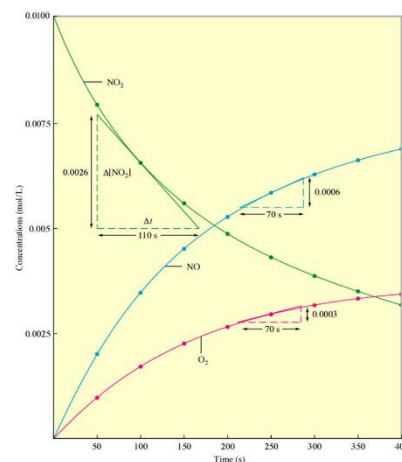
Rate is expressed as _____ because $[C]$ increases)



- The rate for the above equation can be EXPRESSED in terms of any of the reactants or products. Each form will include the _____ as well as a +/- term to account for the increase (product) or decrease (reactant) of the species.
- Write expressions for the rate in terms of each species.

Reaction Rate changes over time.

- The reaction rate (change in concentration over time) is not _____, but instead _____ over time as the concentration of reactants _____.
- Compare the slope of the change reactants at 250 seconds. What do you notice about the slopes and the product coefficients?

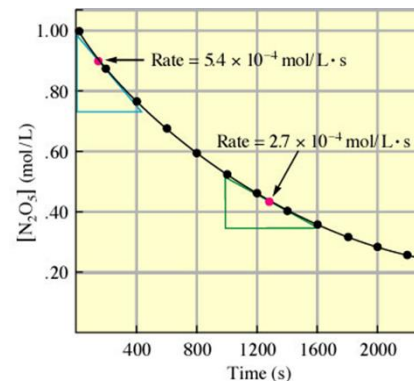


Average rate versus instantaneous rate

_____ Rate is similar to the _____ of concentration vs time function at any two points.

_____ Rate is the _____
_____ to any point on the function- can tell us the rate at any one point in _____.

Rate = -(slope of the tangent line)



a

actual

Differential Rate Law or Rate Equation

For the reaction $aA + bB + \dots \rightarrow gG + hH + \dots$

- Rate = $k[A]^m[B]^n \dots$
- $[A]$ & $[B]$ represent molarities of _____.
- m and n indicate reaction _____ for each reactant
- **Note that $m \neq a$ and $n \neq b$**
- Remember that the rate can be expressed in terms of any reactant or product, so for this equation,

$$\text{Rate} = \frac{-\Delta[A]}{a\Delta t} = \frac{-\Delta[B]}{b\Delta t} = \frac{\Delta[G]}{g\Delta t} = \frac{\Delta[H]}{h\Delta t} \dots\dots\dots$$

Rate = $k[A]^m[B]^n \dots$

- The exponents reflect the _____ of the reactants. The _____ of the exponents is the order of the reaction.
- *Example:* For $R = k[A][B]^2$
-the reaction is _____ order in A, _____ order in B and _____ order ($1+2 = 3$) overall.
-Reactant orders (exponents) can be _____ or _____, _____ or _____.
-usually positive integers (small whole numbers)

- **k = rate constant**
- value depends on reaction, temperature and presence of a catalyst
- faster the reaction, larger the k value (as T increases, k increases; as T decreases, k decreases)
- Units of k depend on _____.
- can be calculated by reaction orders and units of concentration and rate.

Units for k

- The “shortcut” to determining units of k is as follows:
 $k = L^x / (\text{mol}^x \cdot \text{time})$ where x is one less than the order of the reaction.
- If the reaction is _____, $k = L^2 / \text{mol}^2 \cdot \text{time}$.
- If the reaction is _____, $k = L / \text{mol} \cdot \text{time}$
- If the reaction is _____, $k = 1 / \text{time}$

Determining Differential Rate Laws from Experimental Data

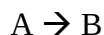
- Differential Rate Law can be observed by _____.
- By varying the _____ of reactants independently, the effect on the rate can reveal the _____ of reaction.
- If _____ the initial concentration of a reactant causes the initial rate to _____, the reaction is _____ order in that reactant.
- If _____ the initial concentration of a reactant causes the initial rate to _____, the reaction is _____ order in that reactant.
- If _____ the initial concentration of a reactant causes the initial rate to _____, the reaction is _____ order in that reactant.

Example 1: Determine the rate law.



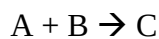
Ex.	[A]	[B]	Rate
1	0.1	0.2	0.10
2	0.1	0.4	0.20
3	0.2	0.4	0.80

Example 2: Determine the rate law.



[A]	Rate
0.05	3×10^{-4}
0.10	1.2×10^{-3}
0.20	4.8×10^{-3}

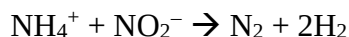
Example 3: Determine the rate law and the value for k with units.



Exp	[A]	[B]	Rate
1	0.10	0.20	1.0×10^{-5}
2	0.20	0.20	1.0×10^{-5}
3	0.20	0.40	2.0×10^{-5}

Example 4: Determine the rate law and the value for k with units.

Ex.	[A]	[B]	Rate
1	0.1	0.1	0.10
2	0.2	0.1	0.20
3	0.4	0.2	0.40

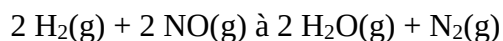
Example 5: Determine the rate law and the value for k with units.

Exp	$[NH_4^+]$	$[NO_2^-]$	Rate(mol/Lmin)
1	0.100	0.005	1.35×10^{-7}
2	0.100	0.010	2.70×10^{-7}
3	0.200	0.010	5.40×10^{-7}

You can use the same method for gaseous reactions with _____!

_____ is directly proportional to _____.

Example: The reaction between H_2 and NO occurs according to the equation



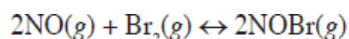
Six trials of the reaction were carried out. The initial rate of change of pressure for each trial was measured and recorded below.

Trial	Initial Pressure (atm)		Initial Rate
	P_{NO}	P_{H_2}	$\Delta atm \ min^{-1}$
I	0.50	0.09	0.025
II	0.50	0.18	0.050
III	0.50	0.27	0.075
IV	0.09	0.80	0.0063
V	0.18	0.80	0.025
VI	0.27	0.80	0.056

Based on these results, what is the rate law for this reaction? Justify your answer.

Multiple Choice Practice

1. The reaction below was performed several times, and the following data was gathered. What is the rate law for this reaction?



Trial	$[NO]_{init}$ (M)	$[Br_2]_{init}$ (M)	Initial Rate of Reaction (M/min)
1	0.20 M	0.10 M	5.20×10^{-3}
2	0.20 M	0.20 M	1.04×10^{-2}
3	0.40 M	0.10 M	2.08×10^{-2}

- (A) Rate = $k[NO][Br_2]^2$ (C) Rate = $k[NO][Br_2]$
 (B) Rate = $k[NO]^2[Br_2]^2$ (D) Rate = $k[NO]^2[Br_2]$

2. A chemical reaction $A \rightarrow 2B + C$ performed with an initial $[A] = 5.25 \times 10^{-3} \text{ M}$ is found to have $[A] = 4.74 \times 10^{-3} \text{ M}$ after 2.5 minutes. What is the concentration of B after 2.5 minutes?

(A) $5.10 \times 10^{-4} \text{ M}$ (B) $9.48 \times 10^{-3} \text{ M}$ (C) $1.02 \times 10^{-3} \text{ M}$ (D) $4. \times 10^{-4} \text{ M}$

3. $A(g) + B(g) \leftrightarrow 2 C(g)$

When the concentration of substance B in the reaction above is doubled, all other factors being held constant, it is found that the rate of the reaction remains unchanged. The most probable explanation for this observation is that

- the order of the reaction with respect to substance B is zero
- substance B is not involved in the reaction
- substance B is probably a catalyst, and as such, its effect on the rate of the reaction does not depend on its concentration
- the reactant with the smallest coefficient in the balanced equation generally has little or no effect on the rate of the reaction

5.3: Concentration Changes over Time

5.4 Elementary Reactions

Integrated Rate Law

- expresses how the _____ of the reactant depends on _____
- instead of _____ initial concentrations and using _____ experiments, a _____ experiment is done and concentration _____ over time are measured.

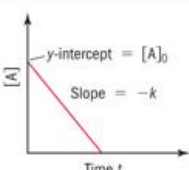
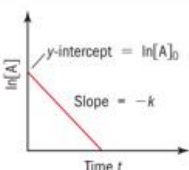
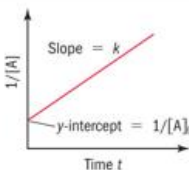
Different orders require different plots to generate _____ lines. Straight line plots are useful for understanding relationships and for making _____

Using Graphs to Determine Order of Reaction

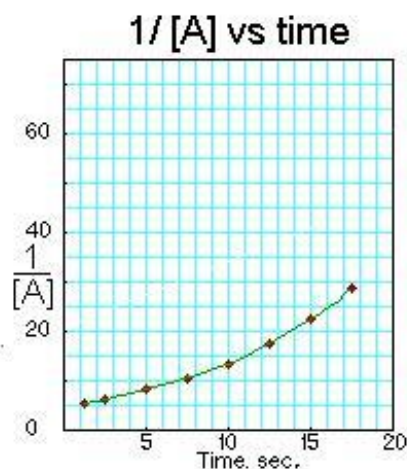
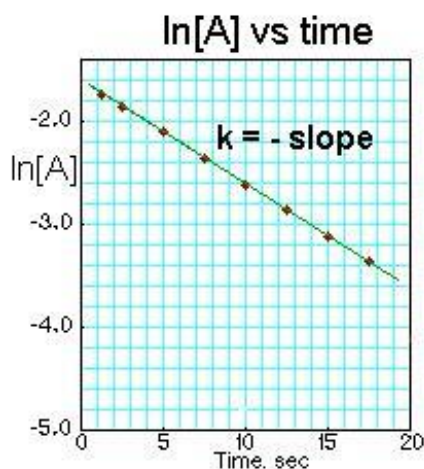
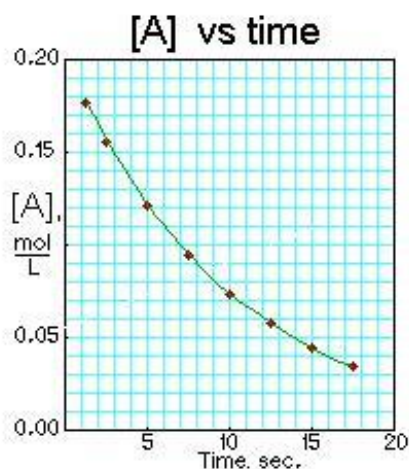
The PLOT of a straight line is your friend! Look for it!

- If plot of $[A]$ versus time is linear, $\text{Rate} = k[A]^0$, the reaction is _____ order for species A.
- If plot of $\ln[A]$ versus time is linear, $\text{Rate} = k[A]^1$, the reaction is _____ order for species A.
- If plot of $1/[A]$ versus time is linear, $\text{Rate} = k[A]^2$, the reaction is _____ order for species A.
- The absolute value of the SLOPE of a linear plot (regardless of order) is the _____.

TABLE 13.2 Rate Law Summary Table

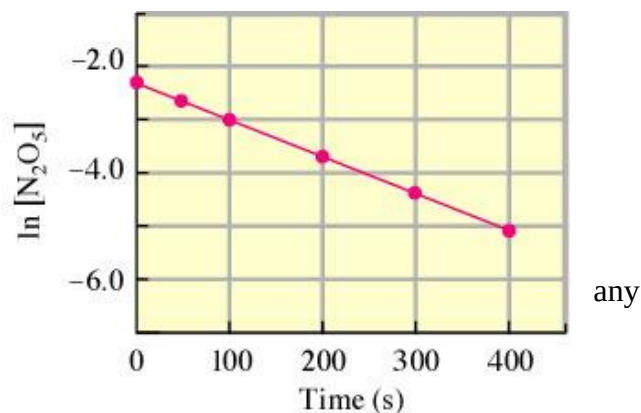
Order	Rate Law	Units of k	Integrated Rate Law	Straight-Line Plot	Half-Life Expression
0	Rate = $k[A]^0$	$M \cdot s^{-1}$	$[A]_t = -kt + [A]_0$		$t_{1/2} = \frac{[A]_0}{2k} = \frac{1}{k} \frac{[A]_0}{2}$
1	Rate = $k[A]^1$	s^{-1}	$\ln[A]_t = -kt + \ln[A]_0$ $\ln \frac{[A]_t}{[A]_0} = -kt$		$t_{1/2} = \frac{0.693}{k} = \frac{1}{k} (0.693)$
2	Rate = $k[A]^2$	$M^{-1} \cdot s^{-1}$	$\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$		$t_{1/2} = \frac{1}{k[A]_0} = \frac{1}{k} \frac{1}{[A]_0}$

What order is this reaction? How do you know?



1st order integrated rate law

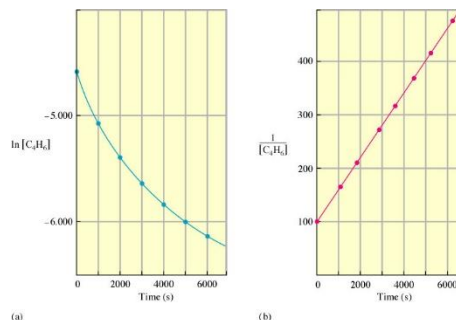
- $\ln [A]_t = -kt - \ln[A]_0$
- A plot of $\ln[A]$ vs t always gives a straight line for a 1st order reaction.
- The slope = $-k$
- This equation can be used to calculate the concentration at given time.



Second order integrated rate law

$$\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$$

A plot of $1/[A]_t$ versus t produces a straight line with slope k .

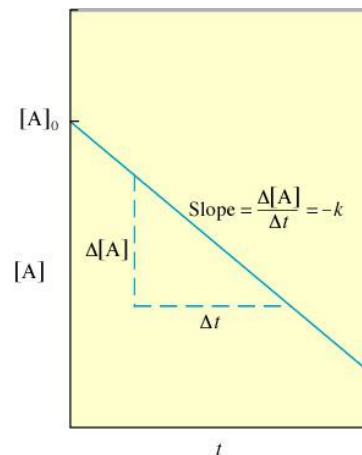


Zeroth order integrated rate law

$$\text{Rate} = k$$

$$[A]_t - [A]_0 = -kt$$

A plot of $[A]$ versus t produces a straight line with slope $-k$.



Multiple Choice Practice!

Questions 1-3 refer to the following reaction and rate laws. $A + B \rightarrow C$

- (A) Rate = $k[A]$
- (B) Rate = $k[A]^2$
- (C) Rate = $k[A][B]^2$
- (D) Rate = $k[A]^2[B]$
- (E) Rate = $k[B]$

1. Which represents a reaction that is zero order with respect to reactant A? _____
2. Which represents a reaction that is second order with respect to reactant A and is third order overall? _____
3. Which represents a reaction that is zero order with respect to reactant B and is second order overall? _____
4. For the reaction whose rate law expression is $\text{rate} = k[X]^2$, a plot of which of the following is a straight line?
 - a) $[X]$ versus time
 - b) $[X]$ versus $1/\text{time}$
 - c) $1/[X]$ versus time
 - d) $\ln[X]$ versus time
 - e) $\ln[X]$ versus $1/\text{time}$
5. As the reaction $A + 2B \rightarrow C$ proceeds at constant temperature, the reaction rate
 - a) remains the same since there is no catalyst present
 - b) remains the same since the temperature is constant
 - c) increases because the rate constant is a large number
 - d) increases because the rate of effective collisions increases over time
 - e) decreases because the concentrations of the reactants decrease as the reaction progresses

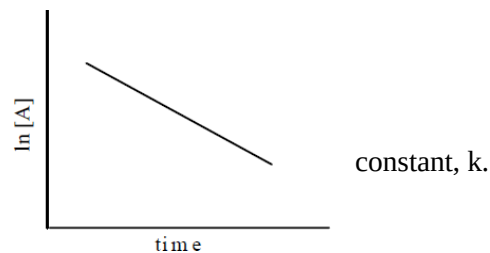
6. Which of the following statements best describe this graph?

I. The graph represents a reactant that is second order.

II. The absolute value of the slope for the graph is equal to the rate

III. The units for the rate constant, k will be $M^{-1}\text{time}^{-1}$

- a) I only
b) I and II only
c) I, II and III
d) II only
e) II and III only



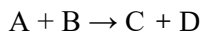
7. For the reaction whose rate law is given below, a plot of which of the following is a straight line?

$$\text{rate} = k[X]$$

- a) $[X]$ vs time
b) $1/[X]$ vs time
c) $\ln[X]$ vs $1/\text{time}$
d) $\ln[X]$ vs time

8. What are the potential units for the rate constant for the reaction below?

- a) s^{-1}
b) $s^{-1} M^{-1}$
c) $s^{-1} M^{-2}$
d) $s^{-1} M^{-3}$



$$\text{rate} = k[A][B]^2$$

9. $2 \text{H}_2\text{O} \rightarrow 2 \text{H}_2 + \text{O}_2$

Which of the following is true of the relative rates of disappearance of the reactants and appearance of the products?

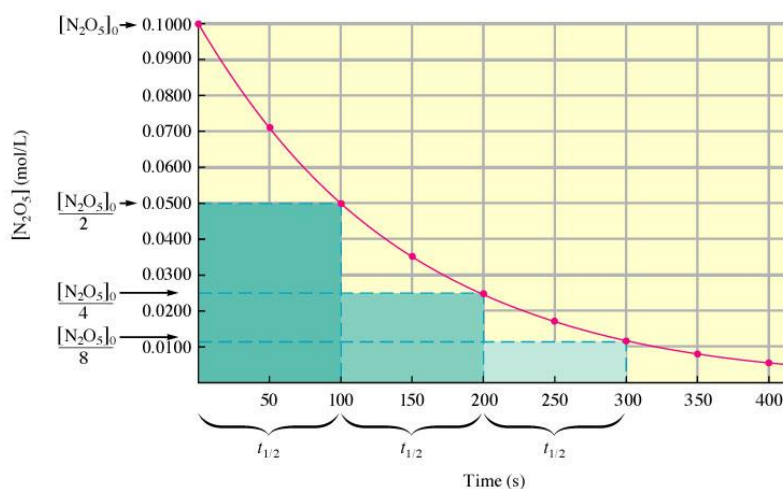
- a) O_2 appears at twice the rate that H_2O disappears.
b) H_2 appears at half the rate that H_2O disappears.
c) H_2 appears at twice the rate that H_2O disappears.
d) O_2 appears at half the rate that H_2O disappears.

Understanding Half Life

- Half-life ($t_{1/2}$) is the length of _____ required for the concentration of a reactant to _____ to _____ of its initial value.
- All reactions have a half life that depend on _____, but _____ order reactions are very unique because their half life value is _____. This means that at any point in a first order reaction, the half life is the same. Any reaction with a _____ half life MUST be a _____ order reaction.

Determine the order of the reaction.

A Plot of (N_2O_5) Versus Time for the Decomposition Reaction of N_2O_5



Calculating half life of a first order reaction

Half life and rate constant are related

- $t_{1/2} = 0.693/k$

A _____ reaction with a _____ $t_{1/2}$ has a _____ k .

A _____ reaction with a _____ $t_{1/2}$ has a _____ k .

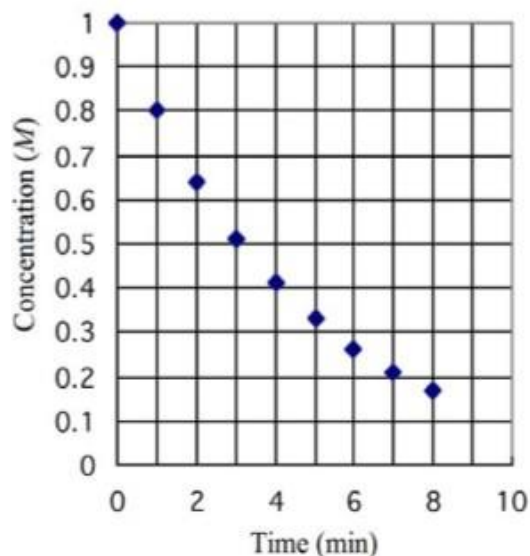
- Common AP Questions involve radioactive decay, which is _____ order.

Example: The decomposition of SO_2Cl_2 and Cl_2 is a first order reaction with a rate constant $k = 2.2 \times 10^{-5} \text{s}^{-1}$ at 320°C . Determine the half-life of this reaction.

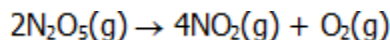
How much reactant remains after 26.25 hours?

Half-Life Multiple-Choice Practice

- Which reaction order is *not* the reaction order of this reaction?
 - 0th order
 - 1st order
 - 2nd order
 - 3rd order
- What is the half-life of the element shown?
 - 2.05 minutes
 - 3.13 minutes
 - 4.18 minutes
 - 5.22 minutes
- What is the molar concentration of this element after a second half-life has elapsed?
 - 0.75 M
 - 0.50 M
 - 0.25 M
 - 0.13 M
- At what time has 75% of the sample decayed?
 - 4.17 minutes
 - 5.33 minutes
 - 6.08 minutes
 - 7.51 minutes



- The following data was collected at 25°C and 1 atmosphere of pressure for the reaction shown below. Which of the following best represents the half-life for this reaction?



Time (minutes)	[N ₂ O ₅] (mol/L)
0	1.24×10^{-2}
10.	0.92×10^{-2}
20.	0.68×10^{-2}
30.	0.50×10^{-2}
40.	0.37×10^{-2}
50.	0.28×10^{-2}
70.	0.15×10^{-2}

- 15 minutes
 - 18 minutes
 - 36 minutes
 - 23 minutes
- After 44 minutes, a sample of ⁴⁴K is found to have decayed to 25% of the original amount present. What is the half-life of ⁴⁴K?
 - 11 minutes
 - 22 minutes
 - 44 minutes
 - 66 minutes

A Brief Summary of Rate Laws

There are _____ types of rate laws!

- ❖ **Differential rate law:** data table contains _____ and _____ data. Compare change in rate when one reactant is held constant but another doubles to determine the order of the reactants and the value of the rate constant, k .
- ❖ **Integrated rate law:** data table contains _____ and _____ data. Use graphical methods to determine the order of a given reactant and the value of the rate constant k .

Integrated Rate Laws Review: set time as your x-axis, and graph three different y-axes (in alphabetical order- CLR!)

Order	Rate Law	Linear Plot	Straight Line Equation	Half-life
0 th	Rate = $k[A]^0 = k$	Concentration, $[A]$ vs time	$[A]_t - [A]_0 = -kt$	totally exists. (but not AP tested)
1 st	Rate = $k[A]^1$	Natural Log, $\ln[A]$ vs time	$\ln[A]_t - \ln[A]_0 = -kt$	$t_{1/2} = \frac{\ln(2)}{k} = \frac{0.693}{k}$
2 nd	Rate = $k[A]^2$	Reciprocal, $1/[A]$ vs time	$\frac{1}{[A]_t} - \frac{1}{[A]_0} = kt$	totally exists. (but not AP tested)

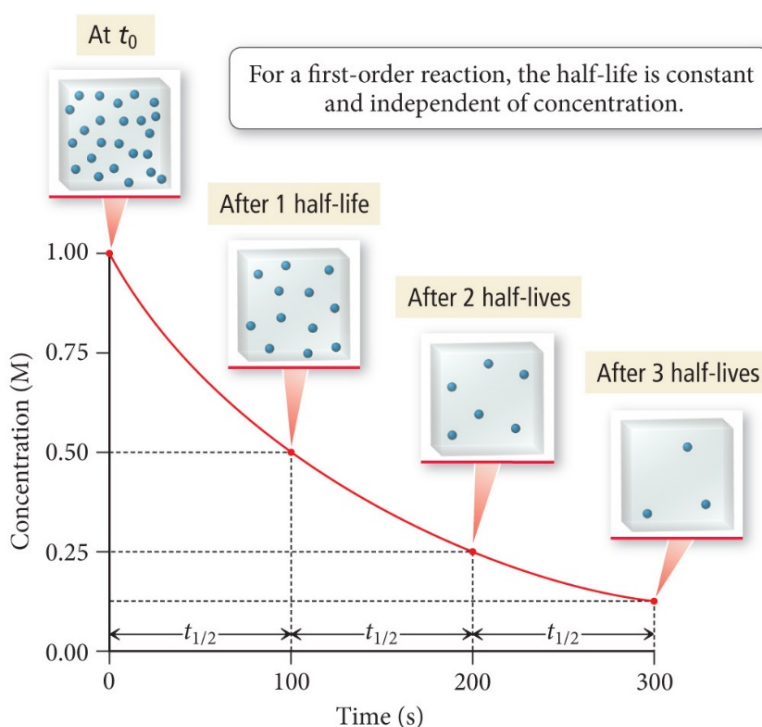
Half-life (_____): time required for _____ of the original reactant sample to disappear.

- ➔ The AP test focuses on the half-life of _____ order reactions only!! They are the only types of reactions for which the length of a half-life is _____.

In a first order reaction, the half-life equation is as follows:

$$t_{1/2} = \frac{\ln(2)}{k} = \frac{0.693}{k}$$

Half-Life for a First-Order Reaction



Let's Practice!

1. A certain first order reaction has a half-life of 20.0 minutes.
 - a. Calculate the rate constant for this reaction.

- b. How much time is required for this reaction to be 75% complete?

2. Dinitrogen pentoxide decomposes according to the equation $\text{N}_2\text{O}_5 \rightarrow \text{NO}_3 + \text{NO}_2$

The following data were collected for this reaction at a given temperature:

$[\text{N}_2\text{O}_5]$	$\ln[\text{N}_2\text{O}_5]$	$1/[\text{N}_2\text{O}_5]$	Time (sec)
1.00	0.00	1.00	0
0.82	-0.20	1.22	25
0.68	-0.39	1.48	50
0.56	-0.59	1.80	75
0.46	-0.78	2.18	100
0.38	-0.98	2.65	125
0.31	-1.17	3.23	150

- a) What is the order of this reaction? Write the rate law expression. Justify your answer (you may include a sketch of a graph as part of your explanation).
- b) Determine the value of the rate constant for this reaction (including units).
- c) Determine the half-life for the reaction under the conditions of this experiment.

Reaction Mechanisms

- Reaction Mechanism- a mechanism of reaction is the step by step _____ of _____ by which overall chemical change occurs.



Write the overall reaction for the above mechanism.

intermediate - a species that is _____ as a product of one step, but _____ as a reactant to a later step

-intermediates are neither _____ nor _____ of the whole reaction

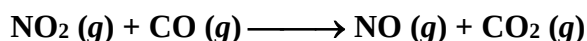
What is the intermediate for the above reaction?

Elementary step - a _____ step in a reaction mechanism, _____ for that step can be written from the balanced step

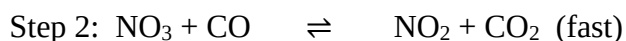
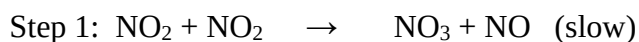
Elementary Reaction - reaction that takes place in _____ - rate law can be written from the reaction

Rate-determining step - _____ step in a reaction, rate of reaction dependent on that step

Mechanism Example



- A proposed mechanism for this reaction is



- How many steps in the mechanism? _____
- Which is the **rate-determining step (RDS)**? _____
- What is the intermediate? Evidence?
- Although CO is necessary for this reaction, it is not involved in the RDS, so the *rate* of the reaction does not depend on its concentration.
- Hypothesize: What *order* will the reaction be for NO₂? For CO?

Molecularity- the number of _____ that must _____ to produce the reaction represented by the elementary _____.

_____ step- a reaction step involving only _____ molecule (rate law always _____ for that elementary step)

_____ step- a reaction step involving the collision of _____ molecules (rate law always _____ for that elementary step)

Reaction Mechanisms Must:

1. Add up to the _____.

2. Agree with the _____ (more about this later)

Note: We can't prove a mechanism absolutely. We can only come up with a possible mechanism based on _____.

5.5: Collision Model

5.6 Reaction Energy Profile

The Collision Model

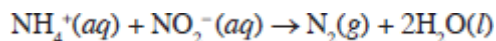
- In a chemical reaction, bonds are _____ and new bonds are _____.
- Molecules can only react if they _____ with each other.
- Furthermore, molecules must collide with the _____ and with _____ to cause bond breakage and formation.

Factors That Affect Reaction Rates (Recall: $R = k[A]^x[B]^y$)- Memorize these!

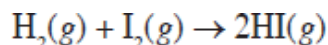
- _____ **of Reactants**- As the concentration of reactants _____, so do the number of _____. (observed in _____)
- _____ **of Reactants**- Different chemicals have different _____ based on their intensive properties. (observed in _____ - changing the reactant will change the k value)
- **Temperature**- At higher temperatures, molecules have more _____, move faster, and collide more often and with greater energy. (observed in _____. Changing rxn temp changes k!)
- **Surface Area**- Chemicals can only react where they _____; greater surface area means greater contact area and more _____. (observed in _____)
- **Catalysts**- Changes speed of the reaction by changing _____. (seen in _____ or entirely new _____)

Let's Practice!

1. Which of the following reasons correctly explains one reason that increasing the temperature of a reaction increases its speed?
 - a. All the reactant molecules will have more kinetic energy.
 - b. A larger percentage of reactant molecules will exceed the activation energy barrier.
 - c. A higher percentage of molecular collisions will have the correct orientation to cause a reaction.
 - d. The order of each reactant will increase.

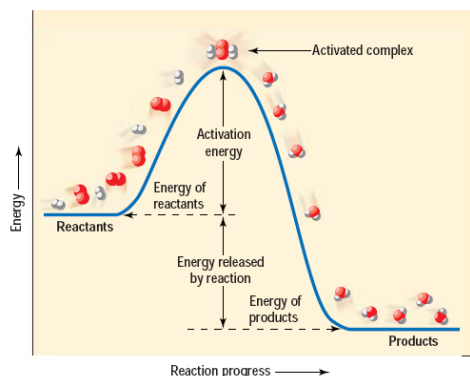


2. Increasing the temperature of the above reaction will increase the rate of reaction. Which of the following is NOT a reason that increased temperature increases reaction rate?
 - a. The reactants will be more likely to overcome the activation energy.
 - b. The number of collisions between reactant molecules will increase.
 - c. A greater distribution of reactant molecules will have high velocities.
 - d. The decreased activation energy is easier for reactant molecules to overcome.



3. When the reaction given above takes place in a sealed isothermal container, the rate law is: $\text{rate} = k[\text{H}_2][\text{I}_2]$. If a mole of H_2 gas is added to the reaction chamber and the temperature remains constant, which of the following will be true?
 - a. The rate of reaction and the rate constant will increase.
 - b. The rate of reaction and the rate constant will not change.
 - c. The rate of reaction will increase, and the rate constant will decrease.
 - d. The rate of reaction will increase, and the rate constant will not change.

Reaction Energy Diagrams



Activation Energy- the energy that must be _____ by the _____ for a successful collision. Collisions with insufficient _____ cannot react.

Activated Complex- a highly _____, _____ particle which immediately _____ to form a product

Enthalpy, or ΔH , or Heat of Reaction: _____ energy absorbed or released during a reaction, measured in _____

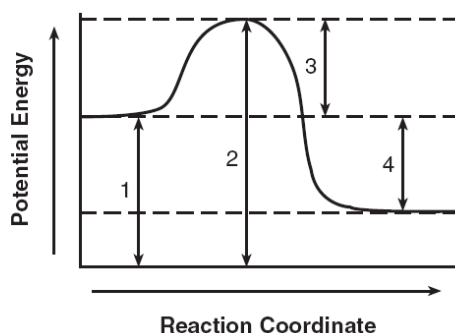
Exothermic Reaction- reaction that _____ energy, products have _____ energy than reactants, _____

Endothermic Reaction- reaction that _____ energy, products have _____ energy than reactants, _____

ΔH is a _____, which means the value _____

_____ on how high the activation energy is, only _____ between the PE of the initial and final states.

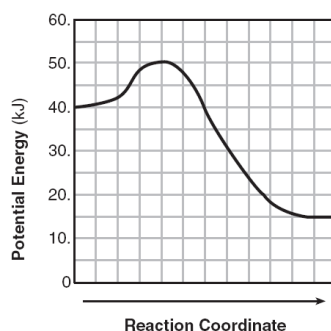
46 Given the potential energy diagram for a reaction:



Which interval on this diagram represents the difference between the potential energy of the products and the potential energy of the reactants?

- (1) 1 (3) 3
(2) 2 (4) 4

43 Given the potential energy diagram for a chemical reaction:



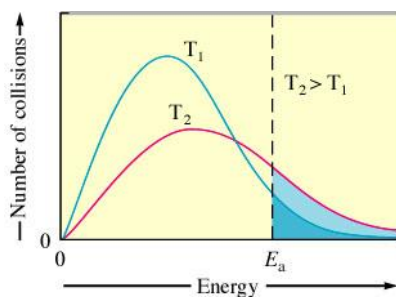
Which statement correctly describes the energy changes that occur in the forward reaction?

- (1) The activation energy is 10. kJ and the reaction is endothermic.
(2) The activation energy is 10. kJ and the reaction is exothermic.
(3) The activation energy is 50. kJ and the reaction is endothermic.
(4) The activation energy is 50. kJ and the reaction is exothermic.

Activation Energy and Catalysts

- Rate of reaction depends on _____, not _____.
- ΔH has _____ on rate of reaction. The _____ the E_A , the _____ the reaction at a given temperature.
- Increasing temperature _____ reaction speed because there will be:
 - More _____ overall
 - More collisions with adequate _____ of collision
- Rate and rate constants often _____ for every 10° increase in temperature. (Rule of thumb)
- Maxwell Boltzmann distributions show this well.

Maxwell Boltzmann Distribution of molecules at 2 different temperatures



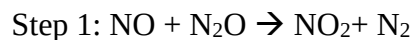
5.7: Intro to Rxn Mechanisms

5.8 Rxn Mechanisms and Rate Law

5.9 Steady State Approximation

Elementary Rxns occur in _____, and because of that, their rate law can be written by _____.

Example: for the *mechanism*:



1. What is the overall rxn?
2. Are there any intermediates?
3. Is there a catalyst?
4. What is the *molecularity*?

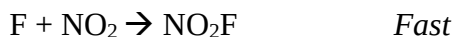
The rate law is **always** written based on the _____ (rate determining step).

Since the RDS for this reaction involves _____ NO molecule and _____ N₂O molecule, it is _____ order in each of these species.

Write the rate law for the reaction.

Multistep Reaction (1st step RDS)

The reaction $2\text{NO}_2(g) + \text{F}_2(g) \rightarrow 2\text{NO}_2\text{F}$ is thought to proceed via the following two-step mechanism:



Is this a possible mechanism for the reaction? How can you tell?

What are the intermediates? Catalysts?

The rate law is written based on the RDS:

$$\text{Rate} = k[\text{NO}_2][\text{F}_2]$$

How could we confirm this rate law?

Compare to the actual rxn. Could the rate law have been derived from the rxn?

Multistep Reaction (non-first RDS)

For a reaction with a 2nd step (or greater) RDS, there may be an _____ as a reactant, and the derivation of the rate law is more complicated.

Example: The reaction $2\text{PO}_2(g) + \text{Cl}_2(g) \rightarrow 2\text{PO}_2\text{Cl}$ is thought to proceed via the following two-step mechanism:



Write the rate law according to the RDS.

Notice the inclusion of an _____ not found in the actual rxn. The rate law *cannot* be expressed in terms of an _____ because the presence of an _____ is unreliable to measure.

Using the Steady State Approximation

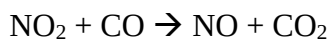
Because the rate law $R = k[\text{P}_2\text{O}_4][\text{Cl}_2]$ contains an intermediate, we must eliminate it using a *steady state approximation*.

The first step is fast and reversible, which means it reaches equilibrium quickly with the forward and reverse rates becoming equal. Write rates for both and set them equal to each other

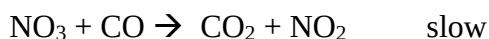
Use algebra to isolate $[\text{P}_2\text{O}_4]$ and substitute into the rate law.

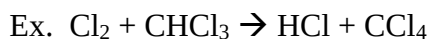
Substituting into our rate law for step 2:

Ex. Find the rate law for the reaction:



With the mechanism:





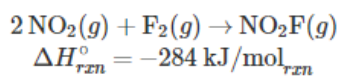
Mechanism



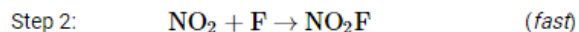
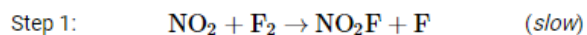
Rate Law:

5.10 Multi Step Rxn Profile

5.11 Catalysis



$\text{NO}_2(g)$ and $\text{F}_2(g)$ can react to produce $\text{NO}_2\text{F}(g)$, as represented above. A proposed mechanism for the reaction has two elementary steps, as shown below.

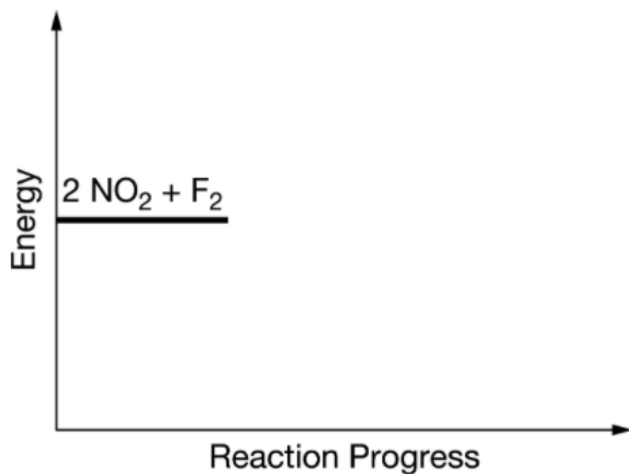


(a) Write a rate law for the overall reaction that is consistent with the proposed mechanism.

(b) On the incomplete reaction energy diagram below, draw a curve that shows the following two details.

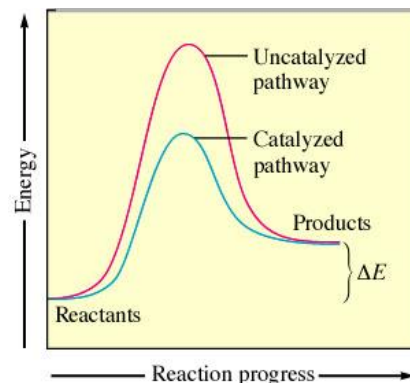
The relative activation energies of the two elementary steps

The enthalpy change of the overall reaction

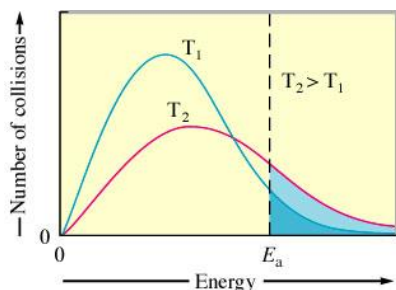


Catalysis

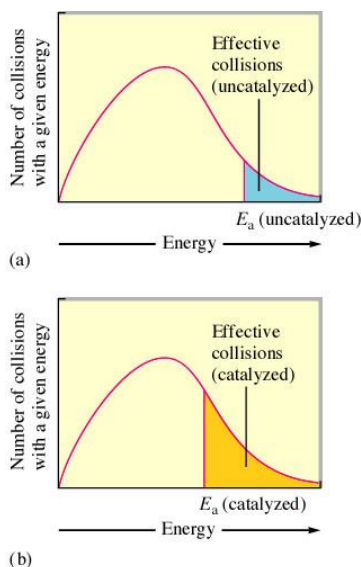
- catalyst- substance that speeds up a reaction without being _____
- Catalysts, such as enzymes in a biological system or the surfaces in an automobile's catalytic converter, _____ the rate of a chemical reaction.
- produces a new _____ with a **lower** _____ **but unchanged ΔH** , _____ (**energy of reaction**)
- A catalyst lowers the E_A for both the forward and reverse reaction. Some catalysts change the _____ of the reaction greatly and others simply change the _____
- Effect on Maxwell-Boltzmann Distribution: Activation Energy (E_a) is _____, allowing for a greater number of _____
- Maxwell Boltzmann Distribution Comparison of _____ vs increase _____ - both increase rate but through different means



Increase in temperature:

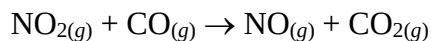


Catalysis:



Multistep Rxn Profiles

Consider the reaction:

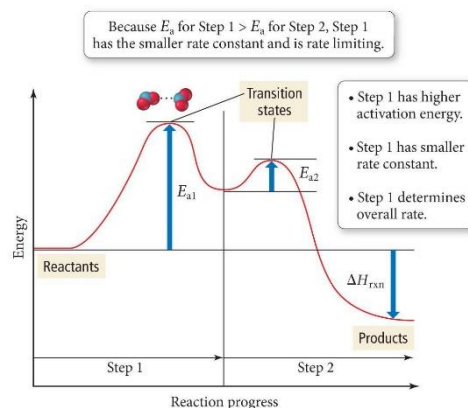


With the experimentally confirmed rate Law:

$$\text{Rate}_{\text{obs}} = k[\text{NO}_2]^2$$

1. $\text{NO}_{2(g)} + \text{NO}_{2(g)} \rightarrow \text{NO}_{3(g)} + \text{NO}_{(g)}$ $\text{Rate} = k_1[\text{NO}_2]^2$ Slow
2. $\text{NO}_{3(g)} + \text{CO}_{(g)} \rightarrow \text{NO}_{2(g)} + \text{CO}_{2(g)}$ $\text{Rate} = k_2[\text{NO}_3][\text{CO}]$ Fast

Energy Diagram for a Two-Step Mechanism

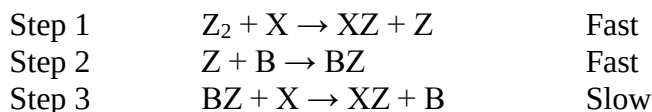


The first step is slower than the second step because its activation energy is larger.

The first step in this mechanism is the rate determining step.

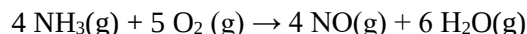
The rate law of the first step is the same as the rate law of the overall reaction.

Example: Given below is the reaction mechanism for a chemical reaction with an enthalpy of $\Delta H = +89 \text{ kJ/mol}$.



- List the overall reaction.
- List any intermediates.
- To the right, draw a rxn energy diagram for the process.

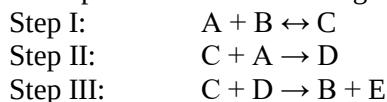
Multiple Choice Practice



- The above reaction will experience a rate increase by the addition of a catalyst such as platinum. Which of the following best explains why?
 - The catalyst causes the value for ΔH to become more negative.
 - The catalyst increases the percentage of collisions that occur at the proper orientation
 - The catalyst introduces a new reaction mechanism for the reaction.
 - The catalyst increases the activation energy for the reaction.

Use the following information to answer questions 2 – 5.

A multi-step reaction takes place with the following elementary steps:



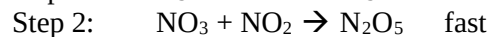
- What is the overall balanced equation for this reaction?

a. $2A + B + 2C + D \rightarrow C + D + B + E$	c. $A + 2C \rightarrow D + E$
b. $A + B \rightarrow D + E$	d. $2A + C \rightarrow E$

3. What is the function of species B in this reaction?
 - a. Without it, no reaction would take place.
 - b. It is a reaction intermediate which facilitates the progress of the reaction.
 - c. It is a catalyst which lowers the activation energy of the reaction.
 - d. It increases the temperature, thus increasing the rate of the reaction.

4. If step II is the slow step for the reaction, what is the overall rate law?
 - a. $\text{rate} = k[\text{A}]^2[\text{B}]$
 - b. $\text{rate} = k[\text{A}][\text{C}]$
 - c. $\text{rate} = k[\text{A}][\text{B}]$
 - d. $\text{rate} = k[\text{A}]/[\text{D}]$

5. Why would increasing the temperature make the reaction rate go up?
 - a. It is an endothermic reaction that needs an outside energy source to function.
 - b. The various molecules in the reactions will move faster and collide more often.
 - c. The overall activation energy of the reaction will be lowered.
 - d. A higher fraction of molecules will have the same activation energy.



6. A proposed reaction mechanism for the reaction of nitrogen dioxide and ozone is detailed above. Which of the following is the rate law for the reaction?
 - a. $\text{rate} = k[\text{NO}_2][\text{O}_3]$
 - b. $\text{rate} = k[\text{NO}_3][\text{NO}_2]$
 - c. $\text{rate} = k[\text{NO}_2]^2[\text{O}_3]$
 - d. $\text{rate} = k[\text{NO}_3][\text{O}_2]$